Data:

Level of the liquid L_1 = 50 ml

Level of the liquid L_2 = 48.5 ml

Level of the liquid L_3 = 51.2 ml

Apparent expansion $= L_3 - L_1$

$= 51.2\,ml - 50\,ml = 1.2ml$

Real expansion $= L_3 - L_2$

$= 51.2\ ml - 48.5ml = 2.7ml$

So, Real expansion > apparent expansion

Example 2

Keeping the temperature as constant, a gas is compressed four times of its initial pressure. The volume of gas in the container changing from 20cc (V_1 cc) to V_2 cc. Find the final volume V_2.

Data:

Initial pressure (P_1)= P

Final Pressure (P_2) = 4P

Initial volume (V_1) = 20cc = $20cm^3$

Final volume (V_2) = ?

Using Boyle's Law, PV = constant

$$P_1V_1 = P_2V_2$$

$$V_2 = \frac{P_1}{P_2} \times V_1$$

$$= \frac{P}{4P} \times 20cm^3$$

$$V_2 = 5\ cm^3$$

TEXTBOOK EVALUATION

T9538

I. Choose the correct answer

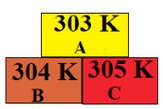
1. The value of universal gas constant
   a) 3.81 $Jmol^{-1}\ K^{-1}$ b) 8.03 $Jmol^{-1}\ K^{-1}$
   c) 1.38 $Jmol^{-1}\ K^{-1}$ d) 8.31 $Jmol^{-1}\ K^{-1}$

2. If a substance is heated or cooled, the change in mass of that substance is
   a) positive b) negative
   c) zero d) none of the above

3. If a substance is heated or cooled, the linear expansion occurs along the axis of
   a) X or −X b) Y or −Y
   c) both (a) and (b) d) (a) or (b)

4. Temperature is the average _____ of the molecules of a substance
   a) difference in K.E and P.E
   b) sum of P.E and K.E
   c) difference in T.E and P.E
   d) difference in K.E and T.E

5. In the Given diagram, the possible direction of heat energy transformation is



a) A ← B, A ← C, B ← C
b) A → B, A → C, B → C
c) A → B, A ← C, B → C
d) A ← B, A → C, B ← C

II. Fill in the blanks:

1. The value of Avogadro number _____
2. The temperature and heat are _____ quantities
3. One calorie is the amount of heat energy required to raise the temperature of _____ of water through_____.
4. According to Boyle's law, the shape of the graph between pressure and reciprocal of volume is _____

III. State whether the following statements are true or false, if false explain why?

1. For a given heat in liquid, the apparent expansion is more than that of real expansion.

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